

Chirping Laser Pulses for Spectral-Focusing CARS

Dispersion, Resolution, and Signal Trade-offs

Justine Major

February 27, 2026

Maîtrise en génie physique

Introduction

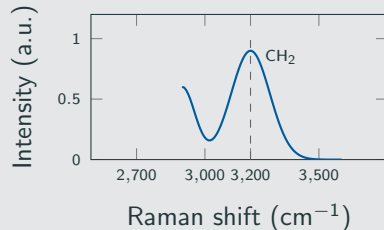
Motivation: Imaging Myelin with Light

Goal: chemically-selective, label-free imaging of myelin sheaths in biological tissue.

- Myelin is rich in **lipid CH₂ bonds**
⇒ strong Raman band near **3200 cm⁻¹**
- Conventional fluorescence: requires exogenous labels, can perturb the sample
- Spontaneous Raman: chemically selective, but signal is *weak* — long acquisition times

Solution: CARS (Coherent Anti-Stokes Raman Scattering)

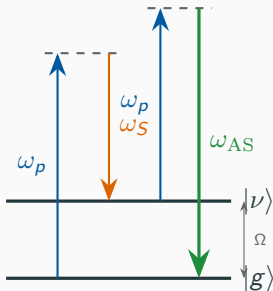
Raman band of interest



Target: $\sim 3200 \text{ cm}^{-1}$ CH₂ stretch

CARS: Four-Wave Mixing Principle

Energy-level picture:



Phase-matching condition:

$$\omega_{AS} = 2\omega_p - \omega_S$$

Resonance condition:

$$\Omega = \omega_p - \omega_S \approx \omega_{\text{Raman}}$$

Signal scales as:

$$I_{\text{CARS}} \propto |A_{\text{Raman}}(\Omega)|^2 \cdot I_p^2 \cdot I_S$$

- Anti-Stokes ω_{AS} is blue-shifted away from excitation $\Rightarrow$ easy to detect

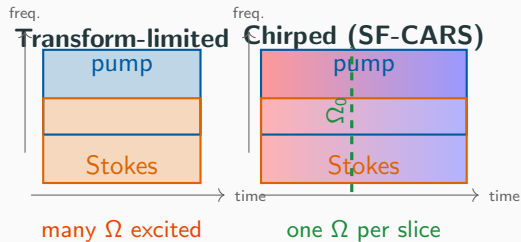
Spectral Focusing: Chirp as a Spectral Filter

Without chirp (transform-limited):

- Both beams are broad in frequency
- All frequency pairs (ω_p, ω_S) overlap simultaneously
- Many Raman modes excited at once
 $\Rightarrow$ poor spectral resolution

With matched chirp (spectral focusing):

- Glass disperses each pulse: each colour arrives at a different time
- At any instant t , only one narrow frequency



Experimental & Simulation Setup

System Parameters

Parameter	Pump	Stokes
Central wavelength	803.31 nm	1041.22 nm
Spectral FWHM (intensity)	11.77 nm	8.92 nm
Bandwidth	2.05 THz	1.80 THz
TL temporal FWHM	80.7 fs	178.9 fs
Time–bandwidth product	0.441	0.441

Glass: S-TIH6

Schott dense titanium flint glass

- High refractive index ($n \approx 1.85$)
- Strong normal dispersion at 800–1050 nm
- Commercially available in thick slabs
- Used here as a *passive chirper*

Two scenarios explored		
	Pump glass	Stokes glass
Scenario A	150 mm	150 mm

Target Raman band

$\Omega \approx 2845 \text{ cm}^{-1}$

S-TIH6 Glass: Dispersion and Chirp Direction

Sellmeier equation for $n(\lambda)$:

$$n^2(\lambda) = 1 + \sum_{j=1}^3 \frac{B_j \lambda^2}{\lambda^2 - C_j}$$

Group index:

$$n_g(\lambda) = n(\lambda) - \lambda \frac{dn}{d\lambda}$$

Since $\frac{dn}{d\lambda} < 0$ (normal dispersion): $n_g > n$

Relative group delay of spectral slice λ_i :

$$\tau_g(\lambda_i) = \frac{d[n_g(\lambda_i) - n_g(\lambda_0)]}{c}$$

Chirp convention

Time domain (at the sample):

Red arrives first $\rightarrow$ Blue arrives last

$\omega(t)$ increases with time

$\Rightarrow$ **positive chirp**

Spatial snapshot (pulse in flight):

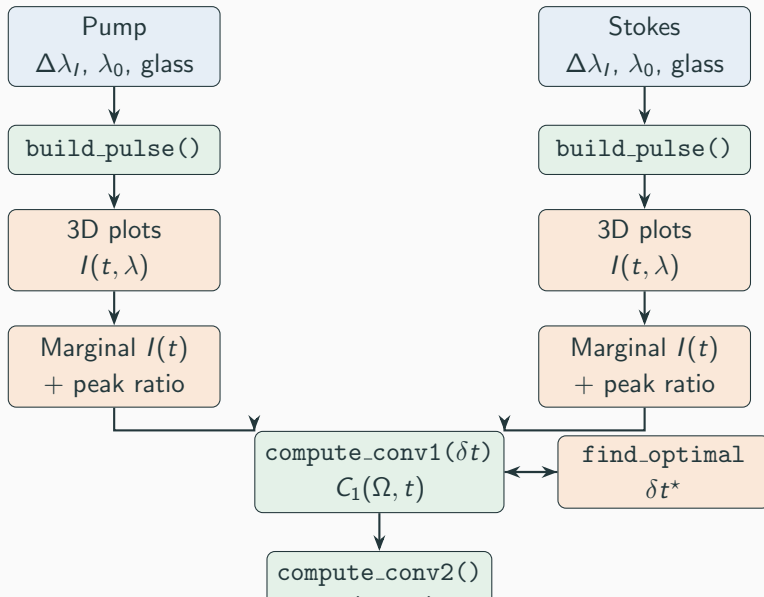
Red at the front (faster)

$\Rightarrow$ "negative chirp" by spatial convention

Same physics, two viewpoints.

This simulation uses the *time-domain*

Simulation Pipeline



Pulse Chirping Through Glass

Physical Convention: Field vs. Intensity

Pulse stored as field amplitude E , not intensity I :

$$E(t, \lambda_i) = A_E(\lambda_i) \exp\left(-\frac{(t - \tau_g(\lambda_i))^2}{\tau_E^2}\right)$$

- $A_E(\lambda_i)$: spectral weight (field Gaussian envelope)
- τ_E : field Gaussian parameter
- $\tau_g(\lambda_i)$: relative group delay of slice λ_i
- Intensity plotted as $I = E^2$

Field vs. intensity FWHM:

$$\text{FWHM}_I = \frac{\text{FWHM}_E}{\sqrt{2}}$$

Why use field amplitudes for convolutions?

Raman coherence is driven by:

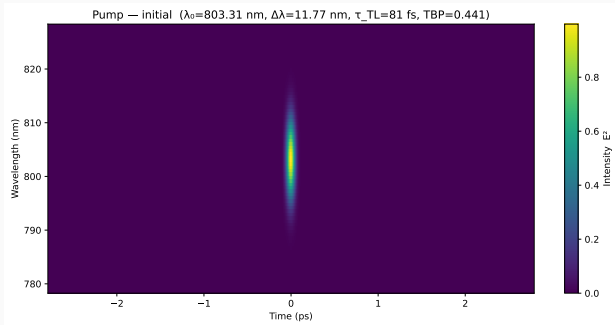
$$\rho(\Omega, t) \propto E_{\text{pump}}(t) \cdot E_{\text{Stokes}}^*(t)$$

This is a *field* $\times$ *field* product — not $\sqrt{I} \times \sqrt{I}$.

Using intensities would be wrong

$I_p \cdot I_S = E_p^2 \cdot E_S^2$ overestimates the effective nonlinear bandwidth by $\sim \sqrt{2}$, artificially narrowing the Raman linewidth.

Pump Pulse — Transform-Limited (Before Glass)



Transform-limited pulse

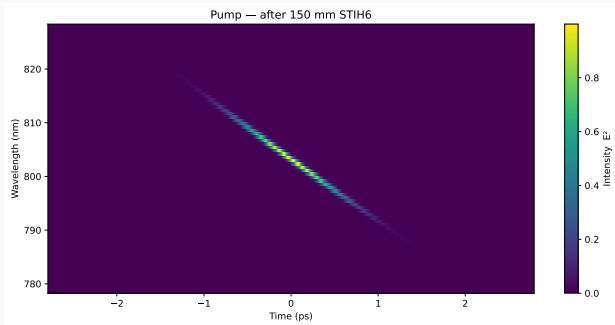
- $\lambda_0 = 803.31$ nm
- Temporal FWHM: 80.7 fs
- All spectral components ($\Delta\lambda = 11.77$ nm) peak *simultaneously* at $t = 0$
- Time-bandwidth product = 0.441 (Gaussian limit)

Reading the plot

The 2D heatmap shows intensity $I(t, \lambda) = E^2$.

At $t = 0$, all wavelengths are bright.

Pump Pulse — After 150 mm S-TIH6 Glass



Chirped pulse — key numbers:

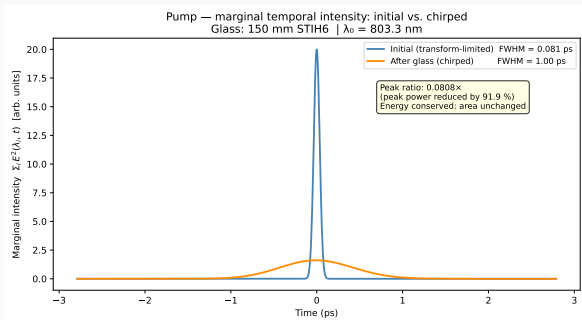
- Group-delay spread: **4.21 ps**
- Elongation: $\approx$ **52 $\times$** the TL duration
- Blue ($\lambda < \lambda_0$): arrives *later*
- Red ($\lambda > \lambda_0$): arrives *earlier*

Reading the plot

The bright ridge is now *tilted*: each wavelength peaks at its own group delay.

This tilt is the **chirp rate** — the linear mapping of frequency onto

Peak Power Reduction: The Musicians Analogy



Marginal temporal intensity $I(t) = \sum_i E^2(\lambda_i, t)$

Why does the peak drop?

Glass is *linear and lossless* — each spectral slice A_k is unchanged. But:

TL pulse: all n_{spec} $\sum_i A_i^2$
slices peak *together* at
 $t = 0$

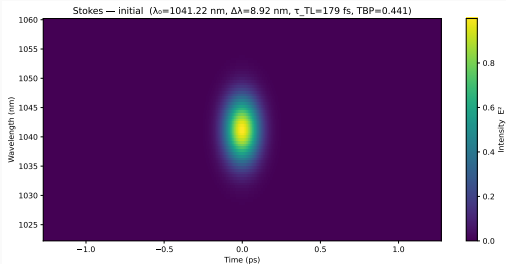
Chirped pulse: slice A_k^2
 k peaks alone at $t = \tau_g(k)$; others negligible

$$\frac{I_{\text{peak}}^{\text{chirped}}}{I_{\text{peak}}^{\text{TL}}} = \frac{\max_k A_k^2}{\sum_i A_i^2} = \frac{1}{n_{\text{eff}}}$$

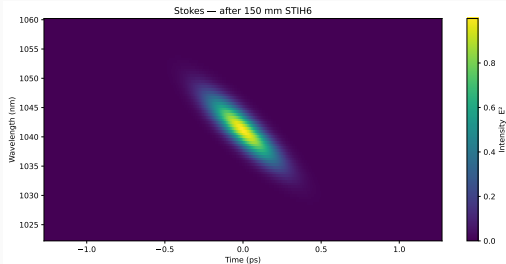
Measured: ratio = 0.080, so

$$n_{\text{eff}} \approx 12$$

Stokes Pulse — Before and After 150 mm Glass



Transform-limited (178.9 fs FWHM)

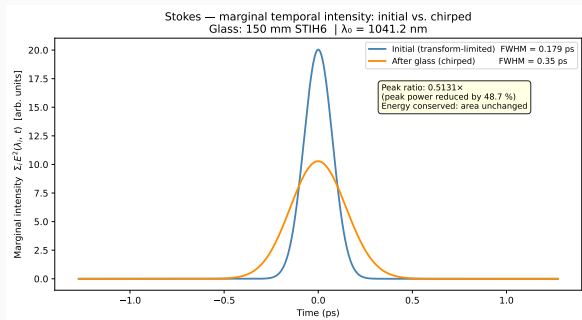


After 150 mm S-TIH6: GD spread = 1.26 ps, elongation $\approx 7\times$

Why is Stokes chirped less than pump?

At 1041 nm the group-velocity dispersion of S-TIH6 is smaller (flatter $n_g(\lambda)$ curve) $\Rightarrow$ less group-delay spread per nm of bandwidth. Same glass thickness, smaller elongation factor.

Stokes Marginal Intensity — Peak Reduction



Stokes (150 mm S-TIH6):

- GD spread: 1.26 ps
- Peak ratio: **0.513** (−49 %)
- Elongation: $\approx 7\times$ TL

Compare to pump:

	Pump	Stokes
GD spread	4.21 ps	1.26 ps
Peak ratio	0.080	0.513
Elongation	$\approx 52\times$	$\approx 7\times$

Pump is much more stretched because it is at a shorter wavelength where S-TIH6 has larger dispersion.

CARS Signal Chain

Raman Excitation — $C_1(\Omega, t)$

The **Raman coherence amplitude** driven by the chirped pulse pair:

$$C_1(\Omega, t) = \sum_k E_{\text{pump}}(\nu_k, t) \cdot E_{\text{Stokes}}(\nu_k - \Omega, t)$$

- Ω : Raman shift (scanned over 2500–3800 cm^{-1})
- ν_k : pump frequency at spectral slice k
- $\nu_k - \Omega$: matching Stokes frequency (CARS condition)
- Sum over all pump spectral slices

Implementation

- Stokes interpolated onto a fine wavenumber grid
- Index arithmetic maps pump ν_k to Stokes $\nu_k - \Omega$
- Vectorised: ~ 10 ms per call
- Scanned over $\delta t \in [-8, +8]$ ps (150 points) to find optimal delay

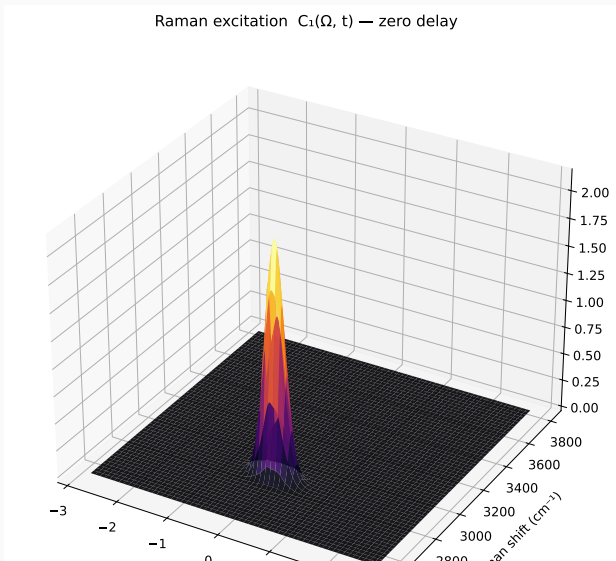
Output

2D array $C_1[\Omega, t]$:

$N_\Omega = 300$ (Raman axis)

$N_t = 1000$ (time axis)

$C_1(\Omega, t)$ at Zero Delay — Raman Coherence Map



What we see:

- Vertical axis: Raman shift Ω (cm^{-1})
- Horizontal axis: time t (ps)
- Color/height: excitation amplitude

Key feature: the ridge of C_1 is approximately horizontal (constant Ω vs. t) for matched chirp rates. This means: at each instant, the same Raman mode is driven, confirming spectral focusing is working.

Finding the Optimal Pump–Stokes Delay

Problem: when chirp rates are mismatched (Scenario B), the centres of mass of the two chirped pulses do not coincide.

Criterion — maximise total C_1 signal:

$$\delta t^* = \arg \max_{\delta t} \sum_{\Omega, t} C_1(\Omega, t; \delta t)$$

This maximises *temporal overlap*, not spectral resolution.

- **Scenario A** (matched, 150/150 mm):
 $\delta t^* \approx 0$ — pulses naturally aligned
- **Scenario B** (mismatched, 50/150 mm):
 $\delta t^* \neq 0$ — pump less stretched, must

What optimal delay is not

It is **not** the delay that minimises the spectral width of $C_1(\Omega)$.

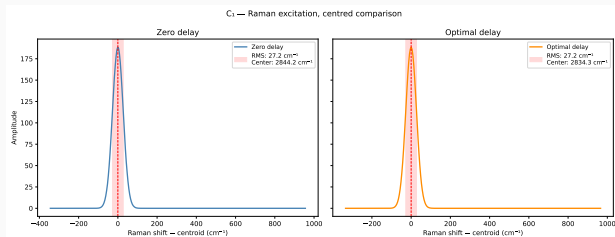
For mismatched chirp rates, the signal-maximising and resolution-optimising delays are different.

Algorithm

Brute-force scan over 150 delay values

$\delta t \in [-8, +8]$ ps
 ≈ 0.6 s total

C₁ Spectral Resolution — Zero vs. Optimal Delay



Scenario A (matched 15 cm / 15 cm):

	$\delta t = 0$	δt^*
RMS width	27.1 cm ⁻¹	27.1 cm ⁻¹
Center	2834 cm ⁻¹	2834 cm ⁻¹

For matched chirp rates, optimal delay ≈ 0 , so the two curves are nearly identical.

Spectral resolution

C₁ RMS width = **27.1 cm⁻¹**

This is the effective Raman linewidth that the system would

Anti-Stokes Signal — $C_2(\nu_{AS}, t)$

The **anti-Stokes field amplitude** generated by a degenerate pump probe reading out the Raman coherence:

$$C_2(\nu_{AS}, t) = \int C_1(\Omega, t) \cdot E_{\text{pump}}(\nu_{AS} - \Omega, t) d\Omega$$

- $\nu_{AS} = \nu_p^{\text{center}} + \Omega$: anti-Stokes frequency
- This is $\omega_{AS} = 2\omega_p - \omega_S$ at field level
- **Output center:**
 $\nu_{AS} \approx 15\,292\text{ cm}^{-1} \approx 654\text{ nm}$
- The probe pump photon carries its own

Physical interpretation

The Raman coherence $C_1(\Omega, t)$ is "read out" by a second chirped pump photon.

Because this probe is *also* frequency-swept, it adds spectral uncertainty to the final CARS signal:

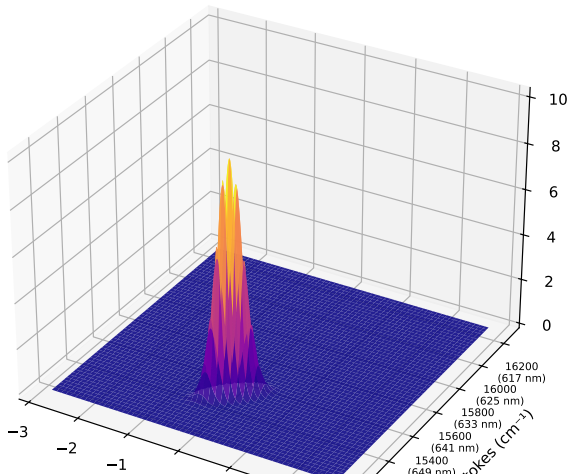
$$\Delta\nu_{AS} > \Delta\nu_{C_1}$$

Implementation

Per-time-step FFT convolution
(`fftconvolve`)

$C_2(\nu_{AS}, t)$ — Time-Resolved Anti-Stokes Signal

sf-CARS signal $C_2(\nu_{AS}, t)$ — optimal $\delta t = -0.054$ ps



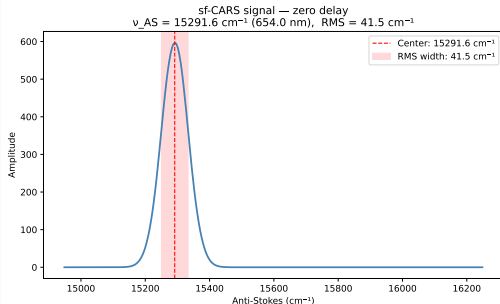
Reading the plot:

- Vertical axis: anti-Stokes ν_{AS} (cm^{-1})
- Horizontal axis: time t (ps)
- Color/height: AS field amplitude
- Center: $\sim 15\,292\text{ cm}^{-1}$ ($\sim 654\text{ nm}$)

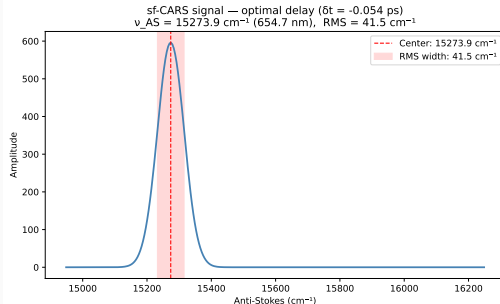
The signal is produced over the entire temporal overlap of the two chirped pulses ($\sim$ ps scale).

Spectral content

C₂ Spectral Projections — Zero vs. Optimal Delay



Zero delay

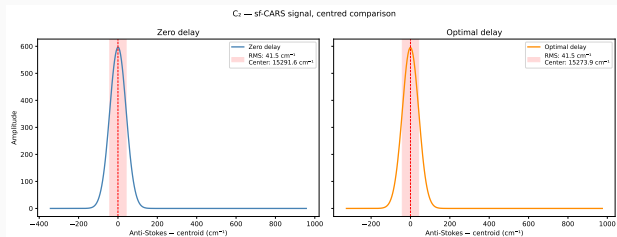


Optimal delay ($\delta t^* \approx 0$ for Scenario A)

Scenario A result: C₂ RMS width = 41.5 cm⁻¹

Broader than C₁ (27.1 cm⁻¹) because the degenerate probe pump photon adds its own spectral uncertainty. Center: 15 292 cm⁻¹ $\approx$ 654 nm (✓ consistent with $\nu_p + \Omega$).

C₂ Final Comparison — Scenario A



Anti-Stokes spectral resolution:

	$\delta t = 0$	δt^*
RMS (cm ⁻¹)	41.5	41.5
Center (cm ⁻¹)	15 292	15 292

For Scenario A (matched chirp), the two delays give essentially identical spectral shapes.

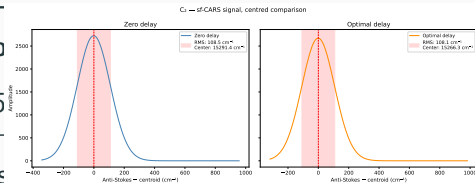
Note

C₂ measures the *anti-Stokes spectrum* as it would appear on a detector. The RMS

Scenario Comparison & Conclusions

Scenario A vs. B — Effect of Pump Glass Thickness

	Scenario A 15 cm / 15 cm	Scenario B 5 cm / 15 cm
Pump GD spread	4.21 ps	1.40 ps
Pump elongation	$\approx 52\times$	$\approx 17\times$
Pump peak ratio	0.080 (−92%)	≈ 0.22 (−78%)
Stokes GD spread	1.26 ps	1.26 ps
Stokes peak ratio	0.513 (−49%)	0.513 (−49%)
C_1 RMS	27.1 cm^{−1}	58.7 cm^{−1}
C_2 RMS	41.5 cm^{−1}	108.5 cm^{−1}



Scenario B: C_2 comparison (RMS = 108.5 cm^{−1})

The Resolution–Signal Trade-off

More glass $\Rightarrow$ better resolution:

- Larger GD spread $\Rightarrow$ longer pulse $\Rightarrow$ slower chirp rate (ps/nm)
- At any instant, a narrower spectral slice is active
- C_1 RMS: 27 cm^{-1} (A) vs. 59 cm^{-1} (B)
— factor **2.2** $\times$ improvement
- C_2 RMS: 41 cm^{-1} vs. 109 cm^{-1}
— factor **2.6** $\times$ improvement

More glass $\Rightarrow$ lower peak signal:

- Pump peak: $0.080\times$ (A) vs. $0.22\times$ (B)

Design rule

For a target Raman linewidth Γ :

$$d_{\text{glass}} \propto \frac{\Delta\nu_{\text{TL}}}{\Gamma}$$

Increasing d beyond what Γ requires wastes peak power without gaining resolution.

Matched chirp rates are essential

Both beams must traverse the *same* glass length so their chirp rates match. Mismatch (Scenario B) degrades resolution

Key Conclusions

1. **S-TIH6 produces positive chirp.** Normal dispersion at 800–1050 nm: blue photons travel slower $\Rightarrow$ frequency sweeps red-to-blue at the sample.
2. **Chirp elongates pulses dramatically.** Pump: 80 fs $\rightarrow$ 4.2 ps (52 $\times$); Stokes: 179 fs $\rightarrow$ 1.3 ps (7 $\times$) through 150 mm glass. Difference due to stronger dispersion at shorter wavelength.
3. **Peak power drops but energy is conserved.** Pump peak ratio = 0.080 = $1/n_{\text{eff}}$ with $n_{\text{eff}} \approx 12$. $\int I(t) dt$ ratio = 1.000 ($\checkmark$).
4. **Matched chirp gives narrowest Raman linewidth.** Scenario A (15/15 cm): C_1 RMS = 27.1 cm $^{-1}$, C_2 RMS = 41.5 cm $^{-1}$.
Mismatched pump (5 cm): $C_1 = 58.7$ cm $^{-1}$, $C_2 = 108.5$ cm $^{-1}$ — factor 2 $\times$ worse.
5. **Degenerate probe broadens the anti-Stokes spectrum.** The second pump photon in C_2 adds spectral uncertainty: $\Delta\nu_{C_2} > \Delta\nu_{C_1}$ by $\sim 50\%$.

Limitations and Future Work

Current limitations:

- **GVD only** — group delay computed from first-order dispersion (group index). Third-order dispersion (TOD) and higher not yet included.
- **Transform-limited input** — assumes input pulses are perfectly unchirped. Real laser outputs have residual chirp.
- **Delay criterion** — optimal delay maximises signal, not resolution. For mismatched scenarios these optima differ.
- **1D spatial model** — no beam profile

Next steps:

1. Add **third-order dispersion** (TOD) to `build_pulse` and verify pulse shape distortion
2. Scan **glass thickness** continuously to produce a resolution vs. signal curve
3. Verify that C_1 ridge is exactly horizontal for matched chirp rates (consistency check)
4. Replace brute-force delay scan with **golden-section search**
5. Extend to **experimental**

Thank you

Questions?

Code: `github.com/` (branch 3d-propagation)

All plots and simulation scripts available.